

Introduction

Good Afternoon, Professor Karam'shetty ([Kuh-rahm-sheh-tee](#)) and Professor Vai'ram ([Vai-rahm](#))!

My name is Jingwen, and for the topic that I'm going to present today, as showed in the slide here is (the) "Efficient supply chain strategy for F&B chains in South East Asia", in particular, examining the ability of **upselling** in the restaurant industry amidst online food delivery growth.

➔ And without further ado, let's begin.

Here's the overall **presentation agenda** for today, we begin with a quick industrial overview, followed by a wide range of literatures review; and from there, we shall see how my research problem and hypotheses were developed.

Then, for the actual empirical analysis part, I'm gonna walk you through from the initial dataset description till the end of result analysis. We finish this presentation by drawing the theoretical findings to practical management applications, as well as some research limitations, and next steps for future investigations.

Restaurant Industry overview

Over the past decade, digital innovation; intensified market competition; and the unforeseen global pandemic have collectively spearheaded a dramatic transformation in the restaurant industry, **which** traditionally dominated by dine-in and takeaway services, has been revolutionised by the 'advent of Online Food Delivery platforms (or **OFD** for short). The global pandemic, outbroke at the end of 2019, acting as an extreme catalyst, fast-tracking the entrenchment of OFD into our daily lives.

The Convenience; Variety; and Safety of online ordering have refined our interactions with restaurants as consumers, and reshaped our '**culinary**' habits in ways that are expected to persist in the long term.

To put this into perspective, the global OFD-sector's market-size has **surpassed 1 trillion** U.S. dollars in 2023, a **dramatic leap** from a forecast of a hundred and fifty-four billion dollars made by the same institution, three years ago.

Literature Review

As we moving through the 'tapestry of existing research, let's pause to consider the key findings and the problem space we are addressing. **Firstly**, Online Food Delivery platforms have been a double-edged sword for the restaurant industry. Their convenience, has indisputably expanded customer bases and **fortified** market presence, but not without cost – such that the commission fees can climb as high as 30% per order, becoming a **heavy-toll ('toe')** on restaurants' profit-margins.

Secondly, the long-term contribution of OFD platforms to profitability remains debatable, given the **razor-thin-margins** within the hospitality industry and its increasingly-crowded market space. (click 'disputable') **This line graph** presents a whole picture on the percentage changes in the revenue growth of online food delivery market over the years.

As we discussed earlier, **pro'pelled by** the increased demand for online deliveries amid the pandemic, Year 2020 saw a remarkable surge, hitting 56% in growth rate. Since that peak, the growing momentum has been moderated, with Year 2022 recording a modest growth of 13%. **|| Looking ahead**, forecasts show a continued positive trend but at a decelerating pace. **||** And in Year 2028, market growth is projected to slow further down, hitting as low as 7%. (Click the green heading)

Hence, the question proposed here is: “How can restaurants adapt and thrive **amidst** such challenges?” The needs for strategic adaptation in customer engagement and sales-tactics are imperative. (And) **Upselling**, the practice of persuading customers to buy-more, or-more-expensive items, is in turn a critical lever (*‘li:ver’*) for profitability in this intensely competitive landscape.

This brings us to the crux of the matter and **the gap** my study aims to fill. While there is a wealth of knowledge on the transformative impact of OFD platforms and the strategic ‘nuances of upselling, there remains a dearth (*‘d3:θ’*) of **empirical** evidence on how these digital platforms specifically affect upselling capabilities within the restaurant sector. -> leading to the identified **problem statement**, my research will delve into this uncharted territory, aiming to investigate the **dynamics** between OFD-platform-reliance and the ability of restaurants to engage in effective upselling strategies.

HypothesisDevelopment

Given the stated problem statement, two opposing hypotheses were developed based on both existing theoretical studies and anecdotal evidence. **On one hand**, we observe that the convenience of online platforms may reduce price sensitivity. The **ease** with which customers can click-and-add items to their orders has revolutionised upselling. Tools that once seemed fu`turistic are now at our fingertips, offering personalised recommendations based on sophisticated algorithms and purchasing behaviours.

These digital capabilities could potentially lead to an increase in the pro`pensity for upselling, providing a customer experience that traditional interactions might not have. These assumptions guide us to the **first hypothesis**, which is: “**Increased dependence on 3rd-party OFD platforms contributes to an enhanced upselling ability**”.

Juxta'posed against this optimistic view is a more cautious perspective. The tactile, human elements that define the dining experience - (e.g.) think of the smell of fresh food, the 'ambiance' of a well-lit dining room, the personal recommendation from a knowledgeable server - may not translate as effectively in the digital realm ('relm').

Furthermore, questions about food quality upon delivery, the absence of direct human interaction, and an increasingly sceptical consumer-base could all lead to a resistance against digital-upselling-attempts. It is this line of reasoning that brings us to a **competing** hypothesis such that: **"Increased dependence on 3rd-party OFD platforms deteriorates restaurants' upselling ability"**.

DatasetDescription

Now, let's move onto the methodo'logical rigour of the study, starting with the description of dataset used in this empirical setting.

The research is powered by a comprehensive dataset, a tapestry of more than 50-Million transactions across a national chain of 99 Quick-Service Restaurants, or QSRs for short, spanning from January to December of the year 2018. A total of roughly 32-thousands-and-7-hundreds datapoints were compiled in this case, representing the daily-operations of these restaurants and offering a rich source of insights into consumer behaviour and sales dynamics.

In particular, the column variables corresponding to each transaction in this dataset could be seen as a narrative - telling us not just the '**what**' and '**how much**' in terms of the sale numbers and the price of Stock- Keeping-Units, and whether the item was sold with discount or engaged with any upselling strategies;

but also the '**when**' and '**how**', which captures the precise time of sale and the mode of order fulfillment, (like) whether it was placed in-store, or pick-up, drive-through, or via an OFD platform.

Variable Constructions and Model Fitting

For the architecture of variables and the formulation of analytical model.

The independent variable in this case is meticulously constructed as a moving average of the platform dependency ratio in a fortnightly-basis (as we can see from the first formula on the top-LeftHandCorner). And its nominator (namely the [PlatformRatio](#)) corresponding to the ratio of sales generated through OFD platforms to the total sales for each QSR on a daily basis. -> ([click 'platformRatio'](#))

As we use a line graph to visualise this ratio and its overall tendency at a median level. What we can observe here is a volatile trend with several spikes and troughs. -> ([click for animation](#))

These fluctuations could be attributed to a multitude of factors, including seasonal demand, promotional activities, and any changes in consumer behaviours. More specifically, **Spikes** could indicate successful promotional campaigns by the delivery platforms or perhaps external events that led to increased reliance on OFD services.

The significant swings in the ratio indicate that the QSR's reliance on platforms **is not constant** but rather responsive to external or operational influences. Such volatility underscores the importance of strategic flexibility and the need for dynamic adjustment to sales strategies within the restaurant industry. -> ([change the slide back](#))

So, by employing a 14-day moving average on this ratio, we're trying to smooth out any daily variations, for example, the chaos of a sudden storm deterring dine-in customers, or a local event driving an unexpected surge in deliveries; therefore, aiming to build a more robust measure that could reflect a stable level of platform reliance.

- ➔ To visualise, imagine a line-chart, where the flow of Platform-Ratio over time creates a story of dependence: a narrative that rises and falls, yet within it, we may identify a consistent trend.

As we can see from this formula on the bottom of this slide, the corresponding parameter for our independent variable is [gamma_1 \(click for animation\)](#), which was estimated through a [logistic regression model \(click for animation\)](#), which showcases the degree of impacts of one unit increase in platform dependence on QSR's upsell ability.

In order to adjust for any cyclical and unique influences that may have any confounding effects on our core variable of interest, a suite of control variables (such as 'day of the week', 'week number', 'public holidays', etc) is also included and demonstrated by the **Matrix X** in our formula. In addition, to mitigate any potential **biases** from unobserved heterogeneity, we've incorporated controls that account for the temporal-trends and promotional campaigns, thereby isolating the pure effect of platform dependency on the upselling capability.

And now, for the dependent variable, which encapsulates the Upselling Ability of each restaurant, was defined in a twofold manner. These ratios, respectively, represent the monetary and quantitative success of upselling within the total sales-landscape, painting a comprehensive picture of upselling 'efficacy. [\(click the 'LogisticGraph'\)](#)

Results and Analysis

After employing the **Generalised Linear Model** to fit a **logistic regression**, and considering a robust set of controls, such as seasonality, holiday periods, and promotional campaigns, the estimated results are presented in this table. These figures are actually **telling** and statistically-significant.

Such that the coefficient for the independent variable on measure1, which reflects the dollar revenue from upselling, is even smaller than -2, with a standard error of 0.45, which indicates a strong negative relationship between platform reliance and upselling revenue. Meaning that: for each 10-percentage-point increase in platform dependence, there will be a nearly **19%** reduction in the proportion of sales **revenue** associated with upselling.

Similarly, with the 2nd measurement, which examines the frequency of upsold transactions, reveals an even more adverse negative relationship. Suggesting more than one-fifth-of a decline in the original **number** of upsold transactions.

➔ And both results stand with high statistical significance, bearing p-values of less than 0.1%.

More importantly, these coefficients are not merely numbers; they are indicators of a critical shift in the operational landscape for restaurants in the age of digital delivery services. What these figures suggest is that an increased reliance on third-party OFD platforms is linked to a discernible deterioration in a restaurant's ability to upsell. This supports our Hypothesis_1, and with a high degree of confidence we are able to reject Hypothesis_0 in this case.

Moderator Analysis

However, from the restaurants point of view, it is not easy for them to forgo the usage of OFD in order to survive in this digital era; therefore, identifying factors that may amplify or buffer this adverse impact is crucial to help business to optimise their engagements with these platforms.

Building on top of a wide range of existing academic research evidence, the following three potential moderating factors, namely customer's 'Affluence level'; 'Party Size'; and 'Peak Mealtime Distribution' were identified.

More specifically, **opulence level** ([click for animation](#)) is measured by the average revenue generated per core menu item transaction. This variable shines a light on the purchasing power of customers frequenting each restaurant and offers insight into how customer wealth may influence upselling potential.

'Party Size' ([click for animation](#)), reflects the number of customers seated together, quantified by calculating the average number of core menu items ordered, serving as a proxy for the group size. As it ranges from intimate tables for two to large gatherings, with the size of the party potentially affecting the likelihood of upselling.

Moving to last factor, which is the **'Peak Mealtime Distribution'** ([click for animation](#)), this metric examines the spread of customer demand throughout the day. By normalising the logarithmic of a lunch-to-dinner transaction ratio, we obtain a clear picture of when restaurants are most likely to encounter opportunities for upselling.

→ Now let's visualise the standardised versions of these factors on histograms.

For 'customers affluence level', the distribution is clearly right-skewed, meaning that there are more data points on the lower end of the affluence scale. The peak of the histogram is around the -0.5 to 0 mark ([click](#)), suggesting that the majority of customers fall within a standard deviation below the mean affluence level. Meanwhile,

the long tail to the right indicates there are fewer customers with high affluence levels, but there is a range present, up to 4 standard deviations above the mean.

For the standardised 'Party Size' factor: unlike the affluence level, this distribution does not exhibit a strong skewness; instead, it presents a bell-shaped curve, symmetrical around the mean. [\(click\)](#) The highest proportion of data points is in the -1 to 1 range, indicating that most party sizes are within one standard deviation of the mean party size. There are fewer instances of extremely small or large party sizes, as shown by the lower bars at the extremes of the histogram.

For the mealtime ratio: its distribution is centred around zero [\(click\)](#), indicating that on average, the number of transactions during lunch and dinner are comparable across the dataset. There is a higher concentration of data points just below zero, suggesting that there are slightly more establishments with a predominant number of transactions during lunch compared to dinner, consistent with the nature of QSRs.

The distribution is not symmetric; showing a skew towards the left, meaning that more restaurants have a higher proportion of lunch transactions than dinner transactions.

ModeratorResult

By adding this highlighted extra term in the previous logistic regression model, the results are fitted and shown in this table. Our major parameter of interest in this case is demonstrated by a vector named '[gamma_3](#)' ([click - here it is](#)), which measures the relationships between upsell ability and those three moderating factors, respectively.

So, for affluence level ([click](#)), the coefficient is 1.76, with a highly significant p-value, suggesting a strong positive relationship, ([click](#)) meaning that wealthier customer segments are likely to enhance upselling revenue, amplifying the negative impact of platform dependence.

Conversely, with a coefficient of -0.1 ([click](#)), which does not present statistical significance given the p-value. ([click](#)) Suggesting that within the scope of our dataset, the number of customers per table does not significantly alter the influence of platform reliance on upselling capabilities.

Turning to Peak Mealtime Distribution ([click](#)), we witnessed a negative coefficient of -0.56, which is also statistically significant, ([click](#)) and intimates those stores with a heavier footfall during lunch hours, as opposed to dinner, are likely to see a greater erosion in upselling revenue with increased reliance on delivery platforms.

Managerial implications

Now, turning our gaze to some of the suggested managerial implications in light of the study's findings. Again, the integration with OFD platforms is not without its tribulations, but through strategic adaptation, the challenges may be transformed into opportunities for growth.

Our analysis indicates that one such strategic pivot is to focus on more affluent customer segments ([click the 'icon'](#)), like the segment circled in red.

These consumers tend to exhibit reduced price sensitivity, making them more receptive to upselling - even within digital environments. Restaurants should thus tailor their online offerings and marketing efforts to appeal to this 'demographic, emphasising quality and exclusivity that align with their higher spending power. ([click the 'red circle'](#))

Another strategic move is to optimise operations for peak mealtime distributions. By having a stronger understanding of the ebb and flow of customer traffic can enable restaurants to capitalise on natural upselling opportunities. And deploying targeted marketing and promotions during these peak times, especially through digital channels, restaurants may be able to harness these critical windows to boost their upselling performance.

To summarise...

A key takeaway from the previous analysis is the marked 18.6% drop in upselling revenue and a 20% decrease in the number of transactions with every 10% rise, indicating a diminished capability with increased dependence. However, it is within the nuances of customer behaviours that we find levers for strategic intervention.

Such that, restaurants need to adapt their business models to balance the advantages of OFD platforms with the challenges they pose to traditional revenue streams. Developing sophisticated digital techniques and leveraging data analytics for

personalised marketing could enhance upselling capabilities in the digital ordering environment. Designed tailored marketing and service strategies to cater to more affluent demographics might help offset some of the negative impacts observed with increasing reliance. [\(Next Page\)](#)

Limitations and Future Investigations

And finally, for some of the **research limitations**: this study primarily relied on information from a proprietary dataset of a QSR chain, which may not universally represent all types of restaurants or global markets.

And secondly, the analysis of customer traits like party composition and meal duration was limited by data availability, potentially overlooking some factors that could influence upselling dynamics.

In addition, external factors such as weather and local events were controlled by statistical means, yet there could be residual impacts that were not fully accounted for, possibly affecting the robustness of our conclusions.

So, the next steps for investigation might focus on exploring a broader array of restaurant types and geographical locations to validate the generalisability of these findings.

Investigating additional customer traits and other potential moderating factors could provide deeper insights into how restaurants can effectively manage upselling strategies in the age of digital food services.

There is also a need for experimental or [longitudinal-studies](#) to better understand **causality** and to examine the long-lasting effects of OFD platforms dependence on restaurant profitability.